

Nathan Fritsche

<https://micronate.com> | nathanf@live.com

Austin, Texas

PRINCIPAL SOFTWARE ENGINEER

Innovative engineering leader combining deep technical expertise with strategic vision to implement generative AI solutions, optimize workflows, and drive digital transformation across complex environments.

- Generative AI Innovator: Pioneer in implementing LLMs, prompt engineering, and AI-driven automation to transform business processes and workflows including multi-modal solutions
- Automation & Innovation Leader: 15+ years of identifying opportunities and building custom solutions that eliminate manual processes and drive operational efficiency
- Prompt Engineering & Context Optimization: Expert in crafting sophisticated prompts, context management, and fine-tuning AI interactions for optimal performance including proactive exploration of GitHub Copilot and similar tools
- Advanced AI Architecture: Design and implement RAG (Retrieval-Augmented Generation) systems, agentic workflows, and MCP (Model Context Protocol) integrations for enterprise-scale AI solutions
- Linux Systems Expert: Deep expertise across major distributions (RHEL, Ubuntu, CentOS, SUSE) with advanced system administration and optimization skills
- Cross-Platform Integration Specialist: Seamlessly bridge Windows and Linux environments, leveraging WSL2, Hyper-V, and hybrid infrastructure solutions
- Containerization & Virtualization: Production experience with Docker, Kubernetes, VMware, and modern container orchestration platforms
- Modern Data Solutions: Design and implement data pipelines using SQL, PostgreSQL, and PowerBI for real-time analytics and business intelligence
- API Architecture Veteran: 15+ years developing and deploying RESTful APIs, microservices, and scalable backend systems

EXPERIENCE

AMD, Austin, TX (Contractor)

November 2024 – Present

AI Engineer

- Spearheading **AI Hardware Testing & Validation** using Windows AI and AMD GPU/NPU solutions, including comprehensive AI Benchmarking and ML pipeline (ONNX, Multimodal, LLM) evaluations, leveraging **ROCm** on both Windows and Linux environments.
- Proactively identified team inefficiencies and engineered multiple internal automation tools, including robust REST API clients utilizing OOP, to streamline workflows, reduce manual effort, and deliver enhanced data analysis and reporting for engineers and technicians.
- Architected and implemented a Proof-of-Concept (POC) RAG (Retrieval Augmented Generation) solution to triage, diagnose, and streamline information collection (WinDbg output, customized CSVs) for troubleshooting, significantly enhancing diagnostic efficiency.
- Pioneered and successfully advocated for the adoption of a structured ticketing system for lab administration, building a functional Jira demonstration that led to organizational approval.
- Developing an "Offline Image Servicing" process for adding MCUs and drivers to offline images, establishing new procedures and scripts for the Windows AI Validation Lab.
- Automating local AI Benchmarking on Systems Under Test (SUTs) via Apex Automation and identifying further RAG opportunities across multiple teams and Business Units.

Technical Proficiencies: GPU Acceleration, DirectML, ONNX, ROCm (Windows and Linux), LMStudio, AnythingLLM, Local AI Procyon, WSL, Docker, Hyper-V, Azure Event Hub, Snowflake, Postgres/SQL, PowerBI.

Dell Technologies, Round Rock, TX

July 2019 – July 2024

Principal Software Engineer (November 2021 - July 2024)

As **Security Maturity Lead**, provided technical leadership and strategic governance to a team of 16 engineers spanning embedded systems and software tool development, ensuring strict compliance with enterprise security standards and product security baselines. Additionally, directed a team of offensive and defensive security testing engineers to develop, automate, and execute comprehensive BIOS, UEFI, and firmware security validation test suites, hardware-level vulnerability assessments, and robust verification processes.

- **Secure Product Lifecycle Governance:** Provided strategic and tactical cybersecurity guidance across the entire product lifecycle—integrating security-by-design principles from initial architectural threat modeling through secure release management and continuous security testing.
- **Hands-on Security Architecture:** Led by example through active, hands-on participation in critical cryptographic design, secure hardware architecture, and complex vulnerability remediation sessions.
- **Security Culture & Capability Building:** Delivered comprehensive technical training and mentorship on secure coding practices, hardware-root-of-trust concepts, and low-level vulnerability identification to upskill and empower the embedded systems and BIOS engineering teams.
- **Embedded & Firmware SME:** Acted as the ultimate technical authority and Subject Matter Expert (SME) for complex embedded systems, BIOS, and hardware security challenges. Conducted end-to-end security assessments, including rigorous risk analysis, architectural threat modeling, penetration testing, coordinated vulnerability disclosure/response, and attack surface hardening.
- **Security Process Optimization:** Identified operational gaps and inefficiencies within the security lifecycle; engineered streamlined, secure workflows and comprehensive compliance documentation. Created high-impact executive presentations to secure stakeholder buy-in and successfully scale new security processes organization-wide.
- **Cross-Functional Security Alignment:** Led cross-functional engineering, product, and IT teams in adopting modernized security processes, providing the necessary training, DevSecOps guardrails, and ongoing support to ensure seamless organizational adoption.
- **AI-Driven Security Innovation:** Conceptualized, developed, and deployed an advanced proof-of-concept (PoC) solution for automated threat modeling and rapid vulnerability response. Leveraged a sophisticated stack combining automation, NLP, custom data scraping frameworks, and generative AI to significantly harden the product security posture, enable proactive risk management, and generate automated compliance reporting via Power BI dashboards.
- **Regulatory Compliance & Supply Chain Security:** Ensured strict compliance with U.S. Government regulations and federal cybersecurity mandates (e.g., NIST SP 800-161, Executive Orders on Cybersecurity) by deeply integrating security best practices into the Secure Software Development Lifecycle (SSDL), implementing automated CI/CD pipeline security gates, and institutionalizing Software Bill of Materials (SBOM) management.

Dell Technologies, Round Rock, TX

Sr. Software Engineer (July 2019 - November 2021)

Responsible for creating a solution to automate BIOS and firmware updates on multiple PCs in lab environments consisting of over 1000 physical computers and included installation scripts to install in multiple countries. Attended daily Scrum meetings and practiced Lean Agile software development through active participation in code reviews and sprint meetings.

- Architected and implemented a comprehensive PC lab automation platform, encompassing BIOS and firmware updates, system imaging, and configuration management.
- Designed and developed robust Python-based automation frameworks, leveraging web services to provide REST APIs for facilitating update automation. Solution included standing up a Microsoft Internet Information Server to provide web service hosting in a production environment.
- Performed routine cross-team collaboration with Dell BIOS Validation team to seamlessly integrate Python automation solutions into existing enterprise framework.
- Approached by OS team to expand solution and integrate QFE update capabilities. Using original solution's OO design, quickly added support for automated QFE detection, updating, and verification, expanding its use and value to the company.
- Provided technical documentation and diagrams for Confluence site content including persuasive presentations to Dell upper-level management.
- Mastery of regular expressions for data extraction, validation, and complex pattern matching using Python's re library.

- Created and maintained cryptographic signing tools and automated digital signing processes for BIOS security compliance.

Technical Proficiencies: C++, Python, .NET, PowerShell, PyTest, Regular Expressions (re), Unittest, Flask, Lean Agile, Kanban, Scrum, Git, Microsoft IIS, Windows PE, Windows Deployment Server, Bitbucket, Confluence, Jira, CCTK, bcdedit, diskpart, DISM, MongoDB, MySQL, UEFI.

NF Computer Services, Austin, TX

2016 - Current

Principal Software Engineer: Senior Consultant

Remaining at the forefront of Generative AI by staying informed of the latest research and applying emerging technologies. Leveraging hands-on professional experience in software development, embedded systems, data analytics, and cloud integration to provide consulting services for sensor design, environmental monitoring, smart cities, and smart energy initiatives.

- Professional Counsel Member at Gerson Lehrman Group (GLG), providing expert guidance on IoT architecture, device connectivity, market trends and data analytics to Fortune 500 clients seeking strategic advantage.
- Championed the use of generative AI for local healthcare company, showcasing its potential to improve healthcare accessibility.
- Developing AI-driven solutions for natural language processing, including sentiment analysis, text summarization, and machine translation.
- Creating intelligent conversational agents that enhance customer service and support through natural language understanding and response generation.
- Developing proof-of-concept designs for addressing critical healthcare needs, particularly in home care and memory care settings, enabling non-intrusive monitoring of patients with Alzheimer's and dementia.
- Demonstrated expertise in prompt engineering, crafting effective prompts to guide AI models towards desired image outputs.
- Developing novel components and architectures for advanced NLP applications, such as crafting prompt engineering strategies and building AI-driven chatbot systems using the OpenAI API and leading frameworks like LangChain.

Technical Proficiencies: C++, Python, .NET, JavaScript; LangChain, Generative AI, Gen AI, Hugging Face, Sentence-Transformers, AWS (SQS, IoT Hub, Lambda, EC2, S3), EvoThings, Roblox; IoT, AI, NLP, Prompt Engineering, GPT3.5, GPT4, Large Language Model, LLM, 802.15.4 (Zigbee, XBee), Software Automation, Scripting (DOS, PowerShell, Bash; Data Analytics, Device Connectivity, Cloud Computing, RAG Solutions (Retrieval Augmented Generation)

Honeywell, Austin, TX

2016 - 2019

Data Architect (Remote)

Provided technical guidance to engineers across Honeywell and Strategic Business Groups (SBGs) in the development of software solutions for the Sentience IoT platform. Key responsibilities included root cause analysis of complex debug traces and optimization of data analytic pipelines. This role demanded a deep understanding of Microsoft Azure and its associated web services and data pipeline infrastructure within the context of a highly scalable distributed architecture.

- Leveraged advanced data mining techniques and regular expression analysis to develop comprehensive dashboards and automated alerts for network and system performance monitoring and intrusion detection.
- Lead data and support engineer responsible for L3 engineering support which included root cause analysis, performance optimization and resolving complex, large scale production issues occurring on the Honeywell Industrial IOT platform.
- Provided comprehensive SQL training to enable the development of accurate and informative platform health reports for executive and stakeholder consumption.
- Provided L3 support assistance for machine learning engineers extracting data from analytics streams.
- Regularly attend tech-bridges over the phone and responsible for resolving any platform issues affecting clients using Splunk Dashboards, App Dynamics, Azure Portal, and other MSFT tools including remote desktop and debugging environmental issues (CLR environment, heap and memory issues, Orleans runtime related, etc.).
- Participated in post-mortem analysis with executives and stakeholders to identify root causes, implement corrective actions, and prevent recurrence.
- Actively participated in Scrum software development and Lean Agile methodologies in a 100% Telecommuting position.

Technical Proficiencies: AppDynamics, Artifactory, Big Data, CI/CD, Octopus, Splunk Enterprise, ForgeRock, Regular Expressions, Swagger, SQL Management Server, Oracle VirtualBox, C#, .NET, PowerShell, GIT, GitHub, Bamboo, Bitbucket, Confluence, Jira, Microsoft Azure (Serverless computing; Data Lakes, Table & Blob storage, MS SQL, Azure Functions, Azure Active Directory, IoT Hub, Web Apps, Cloud Services, IAM, VM, Remote Access)

WellAware, San Antonio, TX

2014 - 2015

Software and Network Engineer

Delivered end-to-end IoT solutions, from data acquisition and processing to advanced analytics, for the oil and gas industry. Driving innovation and business growth through the development of innovative IoT and M2M solutions for the energy sector.

- Developed and implemented innovative software solutions for the oil and gas industry, leveraging Python and object-oriented design to optimize data acquisition, processing, and analysis.
- Implemented Python based Microservices, utilizing object-oriented design, and implementing the AMQP to process data being pushed to the cloud and then placed into the appropriate queue to be further utilized by the company's SAAS application.
- Established robust data communication infrastructure, including satellite connectivity, to ensure reliable data transfer from remote field locations.
- Design and implementation of support methods, processes, and technical documentation to vastly improve the support of oilfield monitoring, in addition to designing POC for company to increase MTBF for sensitive, field installed sensors traveling along rough paths to installation sites.
- Designed and created a software library to parse data streams from a variety of dissimilar RPMA and Cellular based radios to assist in communication and data aggregation for various data science methods running in the cloud.
- Established relationships with key ISVs and IHVs while researching and implementing new hardware solutions to better fit the demands of automating oil and chemical data, including improving hardware reliability, data integrity, security, and remote support.
- Utilized past start-up experience to implement new processes, identifying key strategies and methods for increasing documentation and collaboration between engineering and other divisions in the company.

Technical Proficiencies: Python, Eclipse, PyCharm, Visual Studio, Geo-Stationary Satellite, Cellular, ZigBee, RPMA, Remote Access, AMQP, Git, PowerShell, Bash and experience with Apple notebooks and macOS applications for development and productivity.

Digi international – Austin, TX

2009 – 2014

Device Cloud Systems Engineer (Onsite)

Lead Systems Engineer for company's high-profile customers where I collaborated with customers to define requirements, design architecture to develop customized greenfield applications and proof of concept designs for smart cities, smart energy, healthcare, fleet management and agriculture.

- Engineered and implemented software solutions using C++ and C# for Digi International's XBee and Zigbee RF modules, including developing firmware for Atmel and Silicon Labs chipsets.
- Strong focus on designing innovative sensor hardware for XBee and Zigbee ecosystems, integrating innovative sensors, and leveraging cloud connectivity for real-time data analytics through cloud-based platforms the XBee Ecosystem.
- Created proof of concept software for Digi's ConnectCore family, using NXP i.MX51 chipsets. Skilled in developing and integrating innovative embedded solutions.
- Designed and implemented a Zigbee-based wireless motion sensor with advanced event detection and power management and created an embedded Python application for a cloud-connected Digi device to act as a Modbus Master for PLC communication at dam sites.
- Developed software utilities for diagnosing cellular connectivity issues across Europe and deployed a scalable fleet management solution using Python and multi-threading for real-time tracking, including GPS data, engine status, and network performance.
- Developed a Python-based automation tool for cellular device provisioning with SMS callback functionality and wrote REST and XML-RPC utilities in Python and C# to support complex debugging and customer collaboration.
- Provided multiple onsite sessions including firsthand training for hardware and software engineering teams (C and C++ software and hardware engineers) looking to develop Python based apps for Industrial level IOT (Covidien).

Technical Proficiencies: C, C#, .NET 4.5, Python, SVN, SourceSafe, ConnectCore, WinCE, BSP, Web Services, ZigBee, Modbus; Embedded Systems, ARM, AVR, Atmel, ATMicro, Freescale, Microchip, TI & STMicro; Industrial Automation, Modbus, PLC, Smart Energy, iDigi Cloud Services.

Digi international – Austin, TX
Sr. Application Engineer (Onsite)

2002 – 2009

Led the design and development of embedded systems software for our USB line of environmental sensors and cameras. This included designing USB drivers and user-mode applications using C, C++, C#, and VB.NET.

- **Firmware and Driver Engineering:** Engineered and optimized firmware for Watchport/V2 and Watchport/V3 devices using Microchip PIC and Atmel ATmega microcontrollers, including C and Assembly programming. Developed new firmware, Windows drivers, and DLLs for the Watchport/VE Video Encoder, ensuring high-quality video processing and seamless hardware integration.
- **Software and Application Development:** Designed and developed C++/MFC-based applications and ActiveX controls with DirectShow filters for USB camera control and image processing. Created the Watchport Manager software for USB device monitoring and advanced analytics and built a custom motion detection DLL for Watchport USB video streams.
- **Invented the “AutoLogout” kiosk application** for secure user management, integrating Watchport proximity sensors for walk-up and walk-away event detection with customizable responses. Developed software used in kiosk applications to ensure data security and HIPAA compliance.
- **Created USB WDM drivers** for cameras and video encoders, including Bulk and Interrupt mode communications. Engaged in Windows kernel-level debugging using SoftICE and WinDbg to resolve complex driver issues.
- **Advanced Image Processing and POS Solutions:** Architected software leveraging DirectShow, VFW, TWAIN, and ActiveX for advanced image processing. Developed JavaPOS drivers for point-of-sale systems using Watchport Proximity sensors.
- **Created “SensorServ”,** a pro-active monitoring solution

Technical Proficiencies:

- Assembly, C, C++, MFC, Java, C#, .NET, Java, JavaScript, XML, XSL, Active-X, DirectShow, SoftICE, WDM, WinDbg, USB, RS232/422/485; TTL; logic and protocol analyzers, oscilloscopes.
- WHQL, Java JPOS, DirectShow, TWAIN, Vfw (Video for Windows).
- National Instruments (NI) LabView.

PRIOR ROLES: USB Technical Support Manager - 2000 – 2006 (Digi International - Austin, TX)

EDUCATION

Bachelor of Science (BS) in Computer Science Candidate - (30 college credits remaining)
The University of Texas, Austin, Texas

TRAINING | CERTIFICATION

- Professional Council Member, Gerson Lehrman Group (<https://glginsights.com/>)
- Microsoft Certified Systems Engineer
- Microsoft Certified Professional + Internet (MCP + I)
- Member, Association for Computing Machinery, The University of Texas at Austin Chapter (www.utacm.org)
- Member, Houston Area League of PC Users, Houston, TX (www.hal-pc.org)

Dell Technologies Inspire Award, January 24, 2024

Received recognition for exceptional teamwork and demonstrated commitment by stepping up to handle additional responsibilities during a critical project phase while a team member was on leave.

Dell Technologies Inspire Award, September 29, 2023

Received recognition for collaborative teamwork and valuable contributions during Penetration Testing, ensuring critical security testing deadlines were met.

Dell Technologies Inspire Award, April 30, 2022

Received recognition from the Director of Security Engineering for making valuable contributions to process refinement and enhancing threat modeling practices.